

Case Study on Medical Imaging Errors

Analyzing Causes and Improvements

PRESENTED BY:

A.Pavithra(192424347)

Introduction

What is Medical Imaging?

- Medical imaging is the process of creating visual representations of the human body to diagnose and monitor diseases.

Common Imaging Techniques

- X-ray
- CT Scan
- MRI
- Ultrasound
- PET Scan

Importance

- Early disease detection
- Accurate diagnosis
- Treatment planning
- Monitoring patient recovery

Case Study

Case Study: Missed Lung Nodule in Chest X-ray

Patient Details

- Age: 52 years
- Complaint: Persistent cough and chest pain

Incident

- Initial chest X-ray was reported as normal.
- Six months later, a CT scan revealed a cancerous lung nodule.
- Delay in diagnosis resulted in advanced-stage lung cancer.

Impact

- Delayed treatment
- Increased healthcare costs
- Reduced survival chances
- Emotional stress for the patient and family

Causes of Medical Imaging Errors

Major Causes

- Human error (radiologist oversight)
- Poor image quality
- Equipment malfunction
- Patient movement during scanning
- Heavy workload and fatigue
- Communication gaps between doctors and radiologists
- Inadequate clinical information

Result

- Misdiagnosis
- Delayed diagnosis
- Incorrect treatment

Improvements and Solutions

How to Reduce Imaging Errors

- Use Artificial Intelligence (AI) for image analysis
- Regular calibration and maintenance of imaging equipment
- Double reading by experienced radiologists
- Continuous training and certification
- Standardized imaging protocols
- Better communication among healthcare teams
- Quality assurance programs

Benefits

- Faster diagnosis
- Improved accuracy
- Better patient safety

Conclusion

- Medical imaging plays a vital role in modern healthcare.
- However, imaging errors can lead to delayed diagnosis and serious health consequences.
- By adopting AI technologies, improving communication, maintaining equipment, and providing continuous training.
- Healthcare organizations can significantly reduce medical imaging errors and improve patient outcomes.

THANK

YOU